

## ASTORIA TONGUE POINT, OR, USA

WMO: 994011

Lat: 46.2074N

Lon: 123.7684W

Elev: 2

StdP: 101.30

Time Zone: -8.00 (NAP)

Period: 06-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |            | Humidification DP/MCDB and HR |            |            |            |            |            | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD<br>to 99.6% DB |            | WSF          |
|---------------|-------------|------------|-------------------------------|------------|------------|------------|------------|------------|-----------------------|-------------|-------------|-------------|--------------------------|------------|--------------|
|               |             |            | 99.6%                         |            |            | 99%        |            |            | 0.4%                  |             | 1%          |             |                          |            |              |
|               | 99.6%       | 99%        | DP                            | HR         | MCDB       | DP         | HR         | MCDB       | WS                    | MCDB        | WS          | MCDB        | MCWS                     | PCWD       |              |
| (a)<br>12     | (b)<br>-1.0 | (c)<br>0.5 | (d)<br>N/A                    | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>12.9           | (k)<br>10.7 | (l)<br>11.7 | (m)<br>11.4 | (n)<br>2.3               | (o)<br>350 | (p)<br>0.524 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |            |             |            |             |            | Evaporation WB/MCDB |            |            |            |            |            | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|------------|-------------|------------|-------------|------------|---------------------|------------|------------|------------|------------|------------|----------------------|------------|
|               |                        | 0.4%            |            | 1%          |            | 2%          |            | 0.4%                |            | 1%         |            | 2%         |            |                      |            |
|               |                        | DB              | MCWB       | DB          | MCWB       | DB          | MCWB       | WB                  | MCDB       | WB         | MCDB       | WB         | MCDB       | MCWS                 | PCWD       |
| (a)<br>8      | (b)<br>4.9             | (c)<br>23.5     | (d)<br>N/A | (e)<br>21.4 | (f)<br>N/A | (g)<br>20.0 | (h)<br>N/A | (i)<br>N/A          | (j)<br>N/A | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A | (o)<br>2.1           | (p)<br>240 |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| N/A                             | N/A | N/A  | N/A | N/A | N/A  | N/A | N/A | N/A  | N/A           | N/A  | N/A  | N/A  | N/A  | N/A  | N/A               |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |  |  |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|--|--|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |  |  |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      |  |  |
| (a) 9.1           | (b) 7.6 | (c) 6.7 | (e) -2.6                   | (f) 30.0 | (g) 2.4            | (h) 2.2 | (i) -4.4                                           | (j) 31.6 | (k) -5.8   | (l) 32.8 | (m) -7.2   | (n) 34.1 | (o) -8.9   | (p) 35.6 |  |  |
|                   |         |         | DB                         |          |                    |         |                                                    |          |            |          |            |          |            |          |  |  |
|                   |         |         | WB                         |          |                    |         |                                                    |          |            |          |            |          |            |          |  |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 11.1   | 6.5   | 6.3   | 7.6   | 9.1   | 11.7  | 14.1  | 16.3  | 16.9  | 16.1  | 12.8  | 9.0   | 6.1   |
|                                                                               | DBStd       | 4.56   | 2.31  | 2.59  | 2.32  | 2.04  | 2.13  | 1.86  | 1.74  | 1.61  | 2.18  | 2.20  | 2.76  | 2.97  |
|                                                                               | HDD10.0     | 519    | 110   | 106   | 81    | 39    | 5     | 0     | 0     | 0     | 0     | 3     | 50    | 125   |
|                                                                               | HDD18.3     | 2674   | 367   | 338   | 333   | 277   | 206   | 128   | 68    | 50    | 74    | 172   | 282   | 379   |
|                                                                               | CDD10.0     | 907    | 1     | 1     | 7     | 12    | 59    | 124   | 194   | 214   | 182   | 90    | 18    | 3     |
|                                                                               | CDD18.3     | 19     | 0     | 0     | 0     | 0     | 1     | 2     | 4     | 6     | 7     | 0     | 0     | 0     |
|                                                                               | CDH23.3     | 76     | 0     | 0     | 0     | 1     | 9     | 6     | 16    | 19    | 24    | 1     | 0     | 0     |
|                                                                               | CDH26.7     | 14     | 0     | 0     | 0     | 0     | 2     | 1     | 3     | 5     | 3     | 0     | 0     | 0     |
| Wind                                                                          | WSAvg       | 2.6    | 2.7   | 2.4   | 2.6   | 2.7   | 2.6   | 2.7   | 2.7   | 2.6   | 2.4   | 2.4   | 2.7   | 2.7   |
| Precipitation                                                                 | PrecAvg     | 2089   | 321   | 217   | 237   | 163   | 98    | 73    | 25    | 35    | 71    | 200   | 322   | 328   |
|                                                                               | PrecMax     | 2621   | 667   | 543   | 477   | 312   | 171   | 145   | 92    | 116   | 277   | 435   | 676   | 584   |
|                                                                               | PrecMin     | 1506   | 131   | 27    | 34    | 63    | 14    | 13    | 2     | 1     | 0     | 35    | 96    | 155   |
|                                                                               | PrecStd     | 349    | 108   | 111   | 97    | 67    | 43    | 34    | 20    | 32    | 64    | 99    | 138   | 121   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 13.0  | 14.0  | 17.6  | 19.1  | 24.4  | 23.5  | 26.0  | 25.6  | 26.6  | 20.9  | 15.7  | 15.1  |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               | 2%          | DB     | 11.9  | 11.8  | 13.5  | 14.8  | 18.7  | 20.1  | 22.3  | 22.2  | 22.7  | 18.4  | 14.8  | 12.8  |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               | 5%          | DB     | 10.9  | 10.5  | 12.1  | 13.1  | 15.8  | 18.0  | 20.3  | 20.6  | 20.5  | 16.8  | 13.9  | 11.4  |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               | 10%         | DB     | 9.9   | 9.5   | 10.8  | 11.8  | 14.5  | 16.8  | 19.1  | 19.5  | 19.0  | 15.6  | 12.8  | 9.9   |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 4.3   | 4.5   | 5.2   | 4.9   | 4.6   | 4.3   | 4.8   | 4.9   | 5.4   | 5.1   | 4.4   | 3.9   |
|                                                                               |             | MCDBR  | 5.8   | 6.0   | 6.9   | 7.2   | 7.8   | 6.9   | 7.6   | 7.3   | 8.5   | 6.8   | 4.8   | 5.4   |
|                                                                               |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        |        | 0.331 | 0.318 | 0.335 | 0.366 | 0.388 | 0.368 | 0.352 | 0.358 | 0.346 | 0.336 | 0.326 | 0.321 |
|                                                                               | taud        |        | 2.478 | 2.553 | 2.518 | 2.392 | 2.350 | 2.430 | 2.503 | 2.483 | 2.526 | 2.548 | 2.524 | 2.497 |
|                                                                               | Ebn at Noon |        | 767   | 866   | 898   | 891   | 879   | 897   | 908   | 890   | 873   | 831   | 768   | 737   |
|                                                                               | Edh at Noon |        | 68    | 76    | 90    | 111   | 120   | 112   | 102   | 101   | 89    | 75    | 65    | 61    |
| All-Sky Solar<br>Radiation                                                    | RadAvg      |        | 1.11  | 1.96  | 2.92  | 4.24  | 5.28  | 5.78  | 6.27  | 5.39  | 4.13  | 2.42  | 1.31  | 0.93  |
|                                                                               | RadStd      |        | 0.17  | 0.26  | 0.35  | 0.38  | 0.47  | 0.51  | 0.48  | 0.41  | 0.35  | 0.28  | 0.17  | 0.13  |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (36 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                         | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | +21     |

Nomenclature: See separate page